Bid for Contract

Regional Integrated Transport Application

Client: Regional Transport Authority (Preston–Lancaster Coastal Region)

Bidder: [Your Company / Group Name]

Date: [Insert date]

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1. Executive Summary

We propose to design, build, and deliver a **mobile-first, integrated transport application** that unifies bus, rail, road, and weather information into a single, reliable service for the Preston–Lancaster coastal region. The system prioritises **clarity, resilience, accessibility, and real-time accuracy**, addressing fragmentation in current information provision while avoiding vendor lock-in and advertising-driven UX.

Our solution uses open standards and open data, provides live departures and disruption awareness, and supports safer routing by favouring well-served hubs over fragile connections. Delivery will follow an iterative, demo-driven approach with continuous integration and deployment.

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2. Understanding of the Problem

Public transport information in the region is fragmented across operators and modes, making multi-leg journeys difficult—especially during delays. Existing commercial tools either prioritise single journeys, introduce ads, charge fees, or route passengers via poorly served interchanges.

The authority requires: - Integrated, multi-operator journey information - Real-time updates and disruption handling - Mobile-first usability and inclusivity - Low operational cost and long-term maintainability

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3. Proposed Solution

3.1 Core Features (Initial Release)

- **Live map view** of buses and rail services using OpenStreetMap
- **Real-time departures** for arbitrary bus stops and rail stations
- **Multi-leg journey planning** across operators
- **Disruption-aware routing** (delays, cancellations)
- **Operator-neutral interface** (no ads, no prioritisation)

3.2 Advanced Features (Second Release)

- **Historical delay analysis** to improve routing reliability

- **Weather and road context** for journey planning
- **Hub-biased routing** to reduce stranding risk
- **Accessibility-focused UX** (contrast, font scaling, touch targets)

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4. Technical Architecture

4.1 Frontend

- **Framework:** Next.js (TypeScript)
- **Design:** Mobile-first, responsive UI
- **Styling:** Utility-based CSS for accessibility and consistency
- **Mapping:** Leaflet.js with OpenStreetMap tiles

4.2 Backend

- **Runtime:** Node.js
- **APIs:** REST + streaming consumers for live feeds
- **Real-time feeds:** STOMP subscriptions for rail movement and signalling
- **Caching:** In-memory and persistent caching for rate-limited feeds

4.3 Data Layer

- **Timetables:** Cached static datasets with refresh mechanisms
- **Live data:** Short-TTL stores for GPS and movement data
- **Historical data:** Aggregated storage for analytics and reliability scoring

4.4 Deployment & CI/CD

- Automated testing and builds
- Staged releases for client feedback
- Monitoring and logging for operational insight

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5. Security & Compliance

- Principle of least privilege for services
- Input validation and schema enforcement
- Secure handling of credentials and secrets
- Completion of an industry-standard security self-assessment
- Adherence to data protection and fair-use policies

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6. Project Plan & Timeline

Phase	Weeks	Deliverables
Research & Bid	11–12	Proposal, system outline
Detailed Design	13–14	Architecture, risks, costings

Phase	Weeks	Deliverables
Initial Build	15–17	MVP, live demo
Second Release	18–19	Advanced features demo
Evaluation	20	User study, feedback
Final Delivery	21	Code, demo video
Security Review	22	Security self-assessment

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7. Team & Responsibilities

- **Project Lead:** Coordination, client liaison
- **Backend Lead:** Data ingestion, real-time feeds
- **Frontend Lead:** UX, accessibility, mobile design
- **DevOps & Security:** CI/CD, audits, deployment

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8. Risks & Mitigations

Risk	Impact	Mitigation
Data inconsistency	Medium	Schema validation, cross-referencing
Feed outages	Medium	Caching, graceful degradation
Scope creep	High	Phased delivery, fixed MVP
Performance issues	Medium	Incremental profiling, load testing

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9. Costing (Indicative)

All costs are based on **role-specific UK average day rates** and a **12-week delivery timeline**, aligned with the proposed project plan. A fixed-price model is used to ensure cost certainty for the client.

Assumptions

- Team size: **6 people**
- Project duration: **12 weeks** (60 working days)
- Public-sector, non-commercial delivery (moderate rates)

Team Roles, Expertise Levels & Day Rates (UK averages)

Role	Level	People	Day Rate
Project Manager	Senior	1	£500
Backend Developer	Senior	1	£475

Role	Level	People	Day Rate
Backend Developer	Intermediate	1	£400
Frontend Developer	Intermediate	2	£375
DevOps / Security Engineer	Senior (part-time)	1	£500

Stage-Based Time & Cost Estimate

Stage	Duration	Roles Involved	Cost (GBP)

| Testing, Evaluation & User Feedback | 1 week | All roles | £11,600 | | Deployment, Documentation & Handover | 1 week | PM (S), DevOps (S) | £9,600 |

Stage	Duration	Roles Involved	Cost (GBP)
Requirements & System Design	2 weeks	PM (S), BE (S), FE (I)	£21,500
Architecture & Data Integration	2 weeks	BE (S), BE (I), DevOps (S)	£23,800
Core Application Development (MVP)	4 weeks	FE (I), FE (I), BE (I)	£36,000

| Testing, Evaluation & User Feedback | 1 week | All roles | £11,600 | | Deployment, Documentation & Handover | 1 week | PM (S), DevOps (S) | £9,600 |

Additional Costs

Core System (Included in Base Price)

Item	Cost
Infrastructure & Hosting (12 weeks)	£1,200
Monitoring & Logging	£600
CI/CD & Deployment Tooling	£500
Contingency (10% of core delivery)	£8,900

Optional Advanced Features (Priced Separately)

The following costs apply **only if advanced features are commissioned** beyond the core system:

Item	Basis	Cost
Advanced Feature Development	2 weeks senior-led development	£24,500

Item	Basis	Cost
Extended Data Storage (historical data)	DB + backups	£600
Additional API Usage / Data Feeds	Rate-limited feeds	£500
Advanced Analytics Compute	Batch processing	£900
Extended Testing & Validation	Reliability testing	£700
Advanced Features Optional Total		**£27,200**

| **Core System Fixed Contract Price** | **£96,000** | | **Total Price (Including Optional Advanced Features)** | **£123,200** |

Costs are based on standard industry day rates and open-source infrastructure to minimise long-term operational spend.

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10. Conclusion

This proposal delivers a **realistic, scalable, and user-focused transport application** aligned with the authority's objectives. Our approach balances innovation with reliability, prioritises public value over commercial incentives, and is deliverable within the proposed timeframe and budget.

We look forward to the opportunity to deliver this service.

Signed:

[Name]

[Role]

[Organisation]